Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 13: Select the correct answer from the given options.

1. Arrange the following changes in the decreasing order of heat evolved.

   (A) 100 g water to ice at 50°C

   (B) 50 g water to ice at 0°C

   (C) 100 g water vapour to water at 100°C

   (D) 50 g water vapour to water at 50°C

   (a) BADC (b) CADB

   (c) BDAC (d) CDAB

2. **Assertion (A):** Glucose is highly soluble in water.

   Reason (R): Water has a high value of dielectric constant.

   (a) Both A and R are true and R is the correct explanation of A.

   (b) Both A and R are true, but R is not the correct explanation of A.

   (c) A is true and R is false.

   (d) A is false and R is true.

3. Two substances 'X' and 'Y' are exposed to air for a long time. X forms an anhydrous salt, whereas Y forms a monohydrate and on heating, its anhydrous salt is obtained. Identify X and Y.

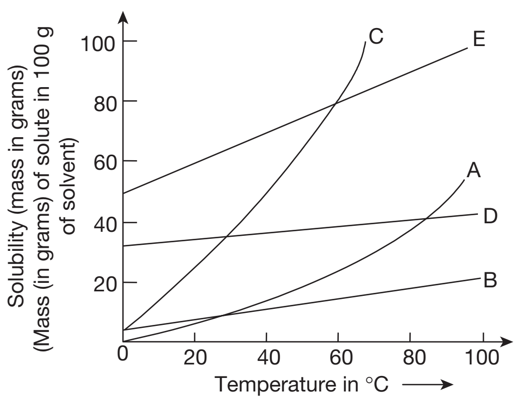
   (a) Blue vitriol, Glauber's salt

   (b) Washing soda, Blue vitriol

   (c) Glauber's salt, washing soda

   (d) Blue vitriol, Green vitriol

4. Based on the following graph, arrange the following in the decreasing order of amounts of substances given below present in dissolved state.



Variation of solubility of various solids with temperature

   (A) B at 10°C (B) A at 70°C

   (C) C at 40°C (D) D at 80°C

Space for rough work

(E) E at 60°C

(a) EBCDA (b) ADCBE

(c) ECDBA (d) ADBCE

5. 80 g of a saturated solution of a substance contains 16 g of the substance. What is the solubility of the substance at that temperature?

(a) 32 (b) 20 (c) 18 (d) 25

6. **Assertion (A):** Glauber's salt shows same solubility at two different temperatures.

Reason (R): Glauber's salt is a hydrated salt.

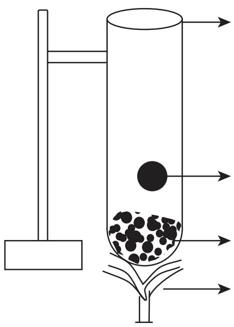
(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true and R is false.

(d) A is false and R is true.

7. What would be the observation of the following experiment?



(a) White dense fumes of P_2O_5 are formed.

(b) Colourless vapours of P_2O_3 are formed.

(c) Vapours of P_2O_5 get condensed on the side of test tube to form fine particles.

(d) There is no reaction.

8. Which of the following applications is not related to the effect of pressure on solubility of gases?

(a) Usage of helium-oxygen mixture by deep sea divers

(b) Preparation of soda water

(c) Difficulty in breathing experienced at higher altitudes

(d) Collection of hydrogen by downward displacement of water

9. 2.1 kJ of heat is supplied to 100 g of water. What would be the expected increase in the temperature of water?

(a) 1°C (b) 0.5°C (c) 10°C (d) 5°C

10. Which of the following samples when taken in equal volumes freezes at the lowest temperature?

(a) Brine solution (b) Saline water

(c) Mineral water (d) Demineralized water

Space for rough work

11. Match the statements of Column A with those of Column B.

Column A (Reaction with Water or Steam)	Column B (Characteristic Features)
(i) Potassium	(A) Reacts only with boiling water
(ii) Iron	(B) Milky alkaline solution
(iii) Calcium	(C) Vigorous and exothermic
(iv) Magnesium	(D) Reversible when carried out in closed container

(a) (i) → (C); (ii) → (B); (iii) → (D); (iv) → (A)
(b) (i) → (D); (ii) → (B); (iii) → (A); (iv) → (C)
(c) (i) → (D); (ii) → (A); (iii) → (B); (iv) → (C)
(d) (i) → (C); (ii) → (D); (iii) → (B); (iv) → (A)

12. When iron metal is taken in place of zinc for the laboratory preparation of hydrogen, the volume of hydrogen collected is found to be lower. Identify the reason.

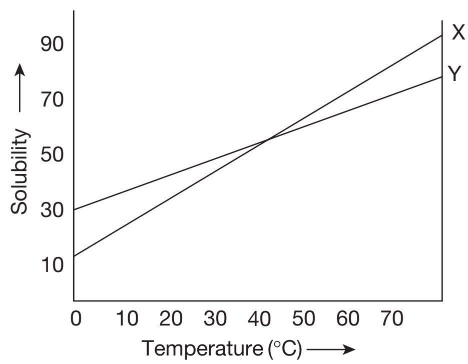
(a) Metal becomes passive due to the formation of protective coating.
(b) Formation of insoluble sulphate stops further reaction.
(c) There are impurities in the metal.
(d) The reaction becomes reversible.

13. Which of the following is not an application of anomalous expansion of water?

(a) Floating of human body in the Dead Sea
(b) Bursting of water bottle filled with water in freezer
(c) Bursting of water pipes in winter season in cold regions
(d) Survival of aquatic animals in frozen lakes

Directions for questions 14 and 15: Answer the following questions based on the solubility curves of 'X' and 'Y' given and select the correct answer from the given options.

14. Saturated solutions of 'X' and 'Y', each dissolved in 100 mL water, are taken at 80°C and subjected to continuous cooling.



Which of the following statements regarding the status of the two solutions at 45°C is true?

(a) X and Y possess the same solubility.

(b) The same amounts of X and Y get precipitated till that temperature.

(c) The same amounts of X and Y would be precipitated on further cooling.

(d) Both (a) and (b)

15. Identify the amounts of 'X' and 'Y' precipitated when the solutions are cooled up to 60°C.

(a) 10 g, 20 g (b) 20 g, 10 g

(c) 60 g, 50 g (d) 50 g, 60 g

Space for rough work

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Which of the following forms represents direct conversion of water vapour into ice?
   (a) Cloud (b) Fog (c) Mist (d) Snow

2. What could be the density of water at 2°C?
   (a) 0.95 g/cc (b) 1 g/cc
   (c) 1.1 g/cc (d) 1.5 g/cc

3. Which of the following samples of water is a bad conductor of electricity?
   (a) Tap water (b) Saline water
   (c) Mineral water (d) Distilled water (freshly prepared)

4. Heat released during the condensation of 1 g of steam at 100°C, to water at 100°C, is utilized to melt x g of ice at 0°C to water at 0°C. Identify x.
   (a) 6.75 g (b) 5 g (c) 5.75 g (d) 6 g

5. Calculate the amount of heat released (in joules) when 10 g of water is cooled from 50°C to 30°C
   (a) 840 (b) 420 (c) 540 (d) 320

6. **Assertion (A):** Water can dissolve many inorganic salts in it.
   Reason (R): Dielectric constant of water is relatively lower than other liquids.
   (a) Both A and R are true and R is the correct explanation of A
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

7. **Assertion (A):** SO_2 is a basic oxide.
   Reason (R): SO_2 forms H_2SO_3 when it is dissolved in water.
   (a) Both A and R are true and R is the correct explanation of A
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

8. Arrange the following metals in the increasing order of the number of H_2 molecules liberated when 1 molecule/atom of the metal reacts with water/steam.
   (A) Na (B) Ca (C) Al (D) Fe
   (a) ABDC (b) CDBA
   (c) DCBA (d) BCDA

9. Arrange the following compartments involved in the purification of water in a proper order.

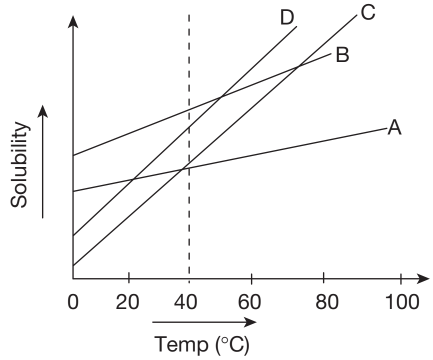
(A) Loading tank (B) Filtration tank
(C) Chlorination tank (D) Sedimentation tank

(a) BCDA (b) CDBA
(c) DABC (d) ADBC

10. Calculate the solubility of the salt A if 25 g of A is present in 225 g of the saturated solution at a given temperature.

(a) 12 (b) 12.5 (c) 1.2 (d) 11.5

11. From the graph given below, identify the amount of salt which is maximum in the same mass of their (A, B, C and D) saturated solutions at 40°C.



(a) A (b) B (c) C (d) D

12. Efflorescent salts are the salts which ___________.

(a) absorb water at room temperature
(b) release water at room temperature
(c) absorb water on heating
(d) release water on heating

13. Solubility of a gas in a liquid __________.

(a) increases with increase in temperature
(b) increases with increase in pressure
(c) does not alter with temperature
(d) does not alter with pressure

14. Identify the composition of water gas among the following.

(a) $C + H_2O$ (b) $CO + H_2O$
(c) $CO + H_2$ (d) $CO_2 + H_2$

15. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Removal of AsH_3, PH_3 from H_2	(A) Lead nitrate solution
(ii) Reaction of Na/K with dilute acid	(B) Reversible
(iii) Reaction of iron with dilute acid	(C) Reaction with sodium
(iv) Removal of H_2S from H_2	(D) Silver nitrate solution
(v) Hydrogen gets reduced	(E) Violent
(vi) Hydrogen gets oxidized	(F) Reaction with chlorine

(a) (i) → (A); (ii) → (B); (iii) → (E); (iv) → (D); (v) → (F); (vi) → (C)
(b) (i) → (A); (ii) → (B); (iii) → (E); (iv) → (D); (v) → (C); (vi) → (F)
(c) (i) → (D); (ii) → (E); (iii) → (B); (iv) → (A); (v) → (F); (vi) → (C)
(d) (i) → (D); (ii) → (E); (iii) → (B); (iv) → (A); (v) → (E); (vi) → (F)

Space for rough work

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. What kind of conversion of state of matter takes place during the formation of dew?
   (a) Water vapour to water (b) Water to water vapour
   (c) Water vapour to ice (d) Ice to water vapour

2. Which of the following statements is correct regarding equal masses of ice and water at 0°C?
   (a) Ice occupies slightly less volume than water.
   (b) Ice occupies slightly more volume than water.
   (c) Ice and water occupy exactly the same volume.
   (d) The volume of ice is half the volume of water.

3. The electrical conductivity of water is due to ____________.
   (a) the presence of dissolved salts
   (b) low dielectric constant
   (c) high specific heat
   (d) high latent heat of vaporisation

4. Compare the mass of ice at 0°C and water at 100°C which can be converted to water at 0°C and steam at 100°C, respectively, when an equal amount of heat energy is supplied to each of them.
   (a) 4:27 (b) 27:4 (c) 5:20 (d) 20:5

5. What is the amount of heat (in joules) required to be supplied to 5 g of water to raise its temperature from 25°C to 75°C?
   (a) 2050 (b) 1050 (c) 1000 (d) 2000

6. **Assertion (A):** Different inorganic salts dissolve in water to a different extent at a particular temperature.
   Reason (R): Dielectric constant of water changes with the nature of salts dissolved in it.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

7. **Assertion:** Metallic oxides react with the base.
   Reason: Metallic oxides form bases when they are dissolved in water.

Space for rough work

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

8. Arrange the following metals in descending order of the number of molecules of water/steam with which one molecule/atom of the metal reacts.

(A) K (B) Fe (C) Al

(a) CAB (b) BCA (c) ACD (d) CBA

9. Arrange the following processes involved in the purification of water in water works in proper order.

(A) Fine sand and gravel are used.
(B) Suspended impurities are allowed to settle.
(C) Chlorine gas is passed through the water.

(a) CAB (b) BAC (c) BCA (d) CBA

10. The solubility of the salt B is 30 at 25°C. Calculate the amount of B present in 390 g of the saturated solution of B at 25°C.

(a) 90 g (b) 30 g (c) 100 g (d) 130 g

11. Solubilities of four different salts is given in the table below. 150 g of each salt is mixed in 750 g of water separately at 25°C.
Identify the incorrect statements.

Name of the salt	Solubility at 25°C
M	25
N	30
O	20
P	15

(a) O is saturated. (b) P is supersaturated.
(c) N supersaturated. (d) M is unsaturated.

12. Which among the following is a deliquescent salt?

(a) Glauber's salt (b) Blue vitriol
(c) Washing soda (d) Hydrated calcium chloride

13. Dissolved gases in water can be expelled by boiling since solubility of a gas in a liquid is ______.

(a) proportional to temperature
(b) inversely proportional to temperature
(c) proportional to pressure
(d) independent of temperature and pressure

14. Which of the following chemical reaction is involved in the Bosch process for the manufacture of hydrogen?

(a) CO is oxidized.
(b) CO_2 is reduced.
(c) H_2O is the reducing agent.
(d) CO is the oxidizing agent.

Space for rough work

15. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Removal of acidic gases present in H_2	(A) Fe catalyst
(ii) Formation of NH_3 from its constituents	(B) Protective coating
(iii) Reaction of Al with dilute acid	(C) P_2O_5
(iv) Removal of water vapour from H_2	(D) Sunlight
(v) Formation of HCl gas from its constituents	(E) Formation of insoluble compounds
(vi) Reaction of Pb with dilute acid	(F) Caustic potash solution

(a) (i) → (F); (ii) → (A); (iii) → (B); (iv) → (C); (v) → (D); (vi) → (E)
(b) (i) → (F); (ii) → (A); (iii) → (B); (iv) → (C); (v) → (E); (vi) → (D)
(d) (i) → (C); (ii) → (D); (iii) → (E); (iv) → (F); (v) → (A); (vi) → (B)
(d) (i) → (C); (ii) → (D); (iii) → (E); (iv) → (F); (v) → (B); (vi) → (A)

Answer Keys

Assessment Test I

1. (b) **2.** (a) **3.** (c) **4.** (a) **5.** (c) **6.** (b) **7.** (a) **8.** (b) **9.** (c) **10.** (b)
11. (c) **12.** (b) **13.** (c) **14.** (c) **15.** (a)

Assessment Test II

1. (d) **2.** (b) **3.** (c) **4.** (c) **5.** (d) **6.** (b) **7.** (d) **8.** (d) **9.** (d) **10.** (a)
11. (d) **12.** (d) **13.** (a) **14.** (a) **15.** (b)

Assessment Test III

1. (d) **2.** (a) **3.** (d) **4.** (a) **5.** (a) **6.** (c) **7.** (d) **8.** (a) **9.** (c) **10.** (b)
11. (b) **12.** (b) **13.** (b) **14.** (c) **15.** (d)

Assessment Test IV

1. (a) **2.** (b) **3.** (a) **4.** (b) **5.** (b) **6.** (c) **7.** (d) **8.** (d) **9.** (b) **10.** (a)
11. (c) **12.** (d) **13.** (b) **14.** (a) **15.** (a)